

**CCS3400 DATABASE PRINCIPLES
FIRST SEMESTER 2025/2026
INDIVIDUAL ASSIGNMENT**

This individual assignment has **2 parts** with the following Learning Outcomes (LO), Student Learning Time (SLT) and marks:

Part	LO	SLT (24 hours)	Marks (20%)
1	Design database using selected data model (C5-CPS1)	12	10
	Draw Entity-Relationship Diagram (ERD) using suitable tools (DS-CPS5)	6	5
2	Create and execute stored procedure and trigger (CTPS-CPS2)	6	5

PART 1 (SLT: 18 hours, Marks: 15%)

Due Date: Week 12 (7 January 2026)

You have been assigned as the team leader for the database design team and is responsible for the conceptual design as well as the logical design for the project.

Description	Output
<u>Conceptual Design:</u> - Create ER model that captures major entities, relationships among entities, and attributes of entities required for a particular system - Create EER model from the extended requirements	- Entity Relationship Diagram - Enhanced Entity Relationship Diagram
<u>Logical Design:</u> Transform the entity/attribute/relationship requirements into relational model for the database and perform normalization if necessary.	Relational database schema

Project Description

AVANTIVE Corporation is a company specializing in the commercialization of automotive parts. AVANTIVE keeps a list of **car models** with information about the **manufacturer, model, and year**. It also keeps several **parts** in stock. A part will have a **part ID, description, unit price, and quantity on hand**. A part can be used for many car models, and a car model has many parts. AVANTIVE has **customers** identified by a **unique customer ID, a name, an address, a phone number, a default shipping address, a date of last purchase, and a date of last payment**. A customer can place many

orders. Each order will have an **order number**, a **date**, a **shipping address**, a **billing address**, and a list of **(order lines)** **part codes**, **quantities**, **list price**, and **extended line totals**. Each order also has a **sales representative ID** (an employee) to identify the person who has made the sale, **an order subtotal**, **an order tax**, **a shipping cost**, **a shipping date**, **an order total cost**, **an order total paid**, and **an order status** (open, closed, or cancel).

- a. Create an ER model to represent the database design. Use suitable tool to draw the Entity-Relationship Diagram (ERD).

The initial project description is extended to include much more detail as follows: AVANTILE Corporation has two types of **customers: retail and wholesale**. All **customers have a unique customer ID**, a **name**, an **address**, a **phone number**, a **default shipping address**, a **date of last purchase**, and a **date of last payment**. **Retail customers have a credit card number**, **credit card type**, **expiration date**, and **email address**. **Wholesale customers have a contact name**, **contact phone number**, **contact email address**, **purchase order number and date**, **discount percentage**, **billing address**, **tax status (if exempt)**, and **tax identification number**. A retail customer **cannot be a wholesale customer and vice versa**. A retail customer normally pays by credit card and is charged the list price for each purchased item. A wholesale customer normally pays via purchase order with terms of net 30 days and is charged a discounted price for each item purchased. (The discount varies from customer to customer.)

AVANTILE Corporation has five departments: administration, marketing, sales, shipping, and purchasing. Each department employs many employees. Each **employee has an ID**, a **name**, a **home address**, a **phone number**, and a **salary and tax ID**. Some employees are classified as **sales representatives**, some as **technical support**, and some as **administrators**. Sales representatives receive a **commission based on sales**. Technical support employees are required to be **certified in their area of expertise**. For example, some are certified as drivetrain specialists, others, as electrical systems specialists. All **administrators have a title and a bonus**.

Introduce enhanced concepts into the ER model designed earlier to accommodate this extension.

- b. Create an EER model from the extended requirements of the database system. Use suitable tool to draw the Enhanced Entity-Relationship Diagram (EERD).
- c. Transform/Map the entity/attribute/relationship requirements into relational model for the database and **perform normalization if necessary**.

PART 2 (SLT: 6 hours, Marks: 5%)
Due Date: Week 13 (14 January 2025)

Continue with the physical design for the project.

Description	Output
<u>Physical Design:</u> • Transform the relational database schema into detail specifications for how to construct actual database using a relational DBMS.	• Technical specification for construction of the database (SQL Script)
• Create stored procedure and trigger for a specific event	• PL/SQL codes with output demonstrating successful execution

- d. Transform the relational database schema into detail specifications for how to construct actual database using Oracle DBMS.
- e. Create stored procedure and trigger for a specific event for the automotive database system.